

Graph Signal Processing and Applications: A Survey

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Abstract—Graph signal processing (GSP) is a field of research with many potential applications, specially in signals with irregular structures. Graphs are structures that represent irregular data attributes. Signal processing approaches mainly focus on analyzing the underlying data. Given the increasing availability of multisensor and multinode measurements, recorded in irregular grids, it can be extremely advantageous to analyze data structured as “signals on graphs”. In this paper, we present a survey on applications and methods to solve problems in different areas of digital signal processing applying GSP.

Index Terms—graph signal processing, irregular structured signal, filtering.

I. INTRODUCTION

Digital signal processing (DSP) is a set of techniques and mathematical concepts with well-known and efficient applications in uniform domains, where signal samples are taken at a constant frequency over time, for example. However, many domains have different structures, that can be modelled as graphs. Graph signal processing (GSP) is an extension of concepts previously developed for traditional signal processing, suited to non-uniform structures.

GSP is an emerging field of research, with an increasing number of scientists approaching specific problems with innovative methods. It has a wide range of applications, such as neuroscience [1]–[3], computer vision [4], image processing [5], swarm intelligence [6] and neural networks [7]. Our objective is to provide a concise survey of some GSP properties and state-of-art applications.

This paper is structured as follows: on Sec. II, we present elementary graph definitions and the Laplacian matrix. On Sec. III, we present Graph Fourier Transforms (GFT) definition. On Sec. IV, we describe a set of GSP operations. On Sec. V we show some state-of-art applications. Conclusions are drawn in Sec. VI.

II. DEFINITIONS

A graph $G = (\mathcal{V}, \mathbf{A})$ is an ordered pair containing a set $\mathcal{V}$ of n vertices and an *adjacency matrix* $\mathbf{A}_{n \times n}$, where $A_{ij} \in \{0, 1\}$. A N -sampled signal is represented by $s_n, N = \{0, 1, 2, \dots, N-1\}$. Rewriting a signal s_n , we obtain a graph signal represented by $\mathbf{s} = [s_0 \ s_1 \ \dots \ s_{N-1}]^T$.

A graph signal $\mathbf{s}$ is a function that maps the elements of $\mathcal{V}$ to a scalar set of real or complex numbers [8], [9]. In this paper, scalars are real numbers and G is undirected. The signal $\mathbf{s}$ can also be seen as a vector $\mathbf{s} \in \mathbb{R}^n$ indexed by the elements of $\mathcal{V}$, such as $\mathbf{s} : \mathcal{V} \rightarrow \mathbb{R} \mid \mathbf{s}(\mathcal{V}_i) = s_i$.

The Laplacian $\mathbf{L}_{n \times n}$ of G is defined in terms of adjacency weights $\mathbf{W}$ and a diagonal matrix of vertices degrees $\mathbf{D}$, as given by

$$\mathbf{L} = \mathbf{D} - \mathbf{W} \quad (1)$$

The matrix $\mathbf{L}$ is symmetric, having n eigenvalues and a set of n mutually orthogonal eigenvectors. The product of $\mathbf{L}$ and a signal $\mathbf{x}$ on the graph G is another signal on the same graph [10], as shown in (2).

$$(\mathbf{L}\mathbf{x})_i = \mathbf{D}_i \left[\mathbf{x}_i - \frac{\sum_j \mathbf{W}_{ij} \mathbf{x}_j}{\sum_j \mathbf{W}_{ij}} \right] \quad (2)$$

The term in brackets informs how much the signal in a given vertex differs from the average signal in neighboring vertices. For a constant signal, $\mathbf{L}\mathbf{x} = 0$. It is equivalent to a homogeneous system with an eigenvalue equal to zero, making the constant signal its respective eigenvector.

Considering the same graph G , an incidence matrix $\mathbf{B} \in \mathbb{R}$ can be used to describe the Laplacian $\mathbf{L} = \mathbf{B}\mathbf{B}^T$, making $\mathbf{L}$ a positive semidefinite matrix, as $\mathbf{x}^T \mathbf{L} \mathbf{x} \geq 0, \forall \mathbf{x}$ [9], [11]. Thus, all eigenvalues λ_l of $\mathbf{L}$ are nonnegative and may be arranged, in order, as elements of a diagonal matrix $\mathbf{\Lambda}$. Since there is an eigenvector $\mathbf{u}_l$ associated to each eigenvalue, the eigenvectors may be arranged as columns of an orthogonal matrix $\mathbf{U}$. Thus, the eigendecomposition of the Laplacian is defined as $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$.

As the Laplacian is directly related to how a signal differs from average values in the neighborhood, its eigenvalues determine how signal polarities change, being analog to frequencies ξ in signal processing [11].

III. GRAPH FOURIER TRANSFORM

A periodic signal may be described as a sum of sinusoids of different frequencies. The lowest frequency that satisfies the condition of periodicity is the fundamental frequency, on which all elements of the harmonic series are based. A periodic signal s is defined by (3), also known as the trigonometric Fourier Series.

$$s(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t \quad (3)$$

where ω_0 is the fundamental frequency, in radians, and coefficients a_n and b_n are the sinusoids amplitudes. The constant a_0 corresponds to the zero harmonic, since $\cos n\omega_0 t = 1$ and

$\sin n\omega_0 t = 0$, for $n = 0$. Euler's formula may be used to convert Fourier Series from trigonometric to exponential form.

$$s(t) = a_0 + \sum_{n=1}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega_0 t} + \sum_{n=1}^{\infty} \frac{a_n + jb_n}{2} e^{-jn\omega_0 t} \quad (4)$$

Considering $a_0 = D_0$ [12] and $D_n = (a_n - jb_n)/2$, Equation (4) is rewritten in a compact expression of the exponential form, often used to describe Fourier Series, as we can see on (5).

$$s(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \quad (5)$$

Coefficients D_n are obtained by an integral over a fundamental period T_0 .

$$D_n = \frac{1}{T_0} \int_{T_0} s(t) e^{-jn\omega_0 t} dt \quad (6)$$

Equations (5) and (6) define exponential Fourier Series and can also be used to represent non-periodic signals. For this purpose, a periodic signal x_{T_0} is built by repeating a non-periodic signal x in intervals of T_0 seconds.

$$\lim_{T_0 \rightarrow \infty} x_{T_0}(t) = x(t) \quad (7)$$

The limit is applied to an exponential Fourier Series of the signal x_{T_0} , representing x when $T_0 \rightarrow \infty$. In this case, we define a function X in terms of the exponential series integral

$$X(n\omega_0) = \int_{-\infty}^{\infty} x(t) e^{-jn\omega_0 t} dt \quad (8)$$

and x_{T_0} may be rewritten as

$$x_{T_0}(t) = \sum_{n=-\infty}^{\infty} \frac{X(n\Delta\omega)\Delta\omega}{2\pi} e^{jn\Delta\omega t} \quad (9)$$

Since $T_0 \rightarrow \infty$ means $\omega_0 \rightarrow 0$, ω_0 is also represented as an infinitesimal $\Delta\omega$. By applying (9) to (7), the signal x is given by (10).

$$x(t) = \lim_{\Delta\omega \rightarrow 0} \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(n\Delta\omega) e^{jn\Delta\omega t} \Delta\omega$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (10)$$

The integral of (10) is known as the Fourier Integral and may be interpreted as an area delimited by the function $X(\omega) e^{j\omega t}$. From (8) and (10), the set of Fourier Transforms (FT) is defined according to (11)

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt &= \mathcal{F}[x(t)] \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega &= \mathcal{F}^{-1}[X(\omega)] \end{aligned} \quad (11)$$

$X(\omega)$ is known as the Direct Fourier Transform of $x(t)$ and $x(t)$ is the Inverse Fourier Transform of $X(\omega)$. Fourier Transforms are useful to specify functions in different domains. Time-domain signals may be represented in the frequency domain, while frequency-domain signals may be reconstituted in the time domain.

Considering the complex exponential $e_{\xi}(t) = e^{2\pi j \xi t}$, the FT is given as an inner product of complex functions [13].

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi j \xi t} dt = \langle f, e_{\xi} \rangle \quad (12)$$

Similarly, a Graph Fourier Transform (GFT) of the signal $\mathbf{x}$ is defined by the eigenvectors of the Laplacian $\mathbf{L}$,

$$\hat{x}_l = \langle x, u_l \rangle \quad (13)$$

Thus, the GFT can be written in matrix notation. The direct GFT is also known as the *analysis equation*, while the inverse GFT is known as the *synthesis equation* [9], [14].

$$\begin{aligned} \hat{\mathbf{x}} &= \mathbf{U}^T \mathbf{x} \\ \mathbf{x} &= \mathbf{U} \hat{\mathbf{x}} \end{aligned} \quad (14)$$

IV. GRAPH SIGNAL PROCESSING

GFT equations are useful for a set of operations commonly used in signal processing. Some examples are filtering, convolution, sampling, translation and others.

A. Filtering

The purpose of filtering is to decompose a signal in Fourier compounds, extract a subset of frequencies and decide which of these to enhance, attenuate or remove [15]. A Fourier Transform (FT) is applied to an input signal, which is multiplied by a filter to generate an output signal. In GSP, graph spectral filtering is defined as

$$\hat{\mathbf{x}}_{out} = \hat{h}(\lambda_l) \hat{\mathbf{x}}_{in}(\lambda_l) \quad (15)$$

B. Convolution

In time domain, convolution between two functions x and g is defined by an integral transform

$$(x * g)(t) = \int_{-\infty}^{+\infty} x(\tau) g(t - \tau) d\tau \quad (16)$$

Conversion to graph spectral domain cannot be done directly because of the term $g(t - \tau)$ [13]. An alternative is to replace complex exponentials with eigenvectors.

$$(x * g)(i) = \sum_{l=0}^{n-1} \hat{x}(\lambda_l) \hat{g}(\lambda_l) u_l(i) \quad (17)$$

C. Sampling and Interpolation

A graph signal $\mathbf{x} \in \mathbb{R}^n$ may be sampled in order to create $\mathbf{x}_M \in \mathbb{R}^m | m < n$, where $\mathcal{M} = (\mathcal{M}_0, \dots, \mathcal{M}_{m-1})$ is the set of sampled vertices. Operator $\mathcal{S}$ performs a linear mapping $\mathbb{R}^n \rightarrow \mathbb{R}^m$.

$$\mathcal{S}_{i,j} = \begin{cases} 1, & j = \mathcal{M}_i \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

Thereafter, $\mathbf{x}_M$ is interpolated by operator $\mathcal{I}$, which performs a mapping $\mathbb{R}^m \rightarrow \mathbb{R}^n$, resulting in signal $\hat{\mathbf{x}} = \mathcal{I}\mathbf{x}_M = \mathcal{I}\mathcal{S}\mathbf{x}$. The eigenvalues of the Laplacian matrix define the bandwidth of graph signals.

D. Translation

Translation operator T may be represented by the convolution of a graph signal $\mathbf{x}$ with a delta centered at a specific vertex v . It works on a kernel $\hat{\mathbf{x}}(\cdot)$, in the spectral domain. The convolution is defined [15] at each vertex i as

$$(T_v \mathbf{x})(i) = (\mathbf{x} * \delta_v)(i) = \sum_{l=0}^{n-1} \hat{\mathbf{x}}(\lambda_l) u_l(i) u_l(v) \quad (19)$$

where the magnitude of $T_v \mathbf{x}(i)$ is reduced as the distance between vertices i and v increases [13].

E. Modulation

Modulation can be interpreted as convolution in the frequency domain, or pointwise multiplication in time or graph domains. The classical operator M_ξ is given by

$$(M_\xi x)(t) = e^{2\pi j \xi t} x(t) \quad (20)$$

The operator may be defined in graph domain by replacing the complex exponential in (20) by a Laplacian eigenvector $(M_l x)(i) = u_l(i) x(i)$. If a multiplication between a low-pass signal and a Laplacian eigenvector u_l is performed, the FT of the resulting vector is supported near the corresponding λ_l eigenvalue [10].

F. Wavelets and Dilation

A wavelet is a finite-energy oscillation capable of decomposing signals into scale components. In classical theory, a mother wavelet ψ is chosen and different versions ψ_s are obtained by scaling (or dilating) ψ

$$\psi_s(t) = \frac{1}{s} \psi\left(\frac{t}{s}\right) \quad (21)$$

where $s > 0$ is a scale factor. Since ψ_s are designed to extract features of a signal x , the inner products of x and each ψ_s are also considered [16]. The Fourier transform of the convolution $y_s = \psi_s * x$ results in

$$\hat{y}_s(\xi) = \hat{\psi}(s\xi) \hat{x}(\xi) \quad (22)$$

The wavelet ψ is a bandpass filter, having no component at $\psi = 0$. The complete wavelet transform is obtained after the addition of the scaling low-pass filter ϕ and the

convolution $y_0 = \phi * x$, which encodes information lost by wavelets ψ_s [10]. Thus, the complete wavelet transform of x is given by the pair $\{(\psi_s * x), (\phi * x)\} = \{y_s | s \geq 0\}$. In graphs, the variable t of (21) represents a vertex, but the meaning of the relation t/s is unclear. The definition of a graph wavelet transform is centered at two functions, $\lambda \rightarrow g(\lambda)$ and $\lambda \rightarrow h(\lambda)$, behaving as bandpass and low-pass filters, respectively. After adjusting (22) to graph signals, the components of y may be defined on frequency domain in terms of eigenvalues λ_l [10].

V. APPLICATIONS

A. Neuroscience

In neuroscience [1], there is a proposed framework to study brain connectomes and to predict autism spectrum disorder (ASD). It is based on two types of connectivity, Functional (FC) and Structural (SC). FC is related to the correlation of Blood-Oxygen-Level Dependent (BOLD) signals, measured by resting-state functional Magnetic Resonance imaging (rs-fMRI). SC is associated to the detection of brain white matter pathways. The authors modelled a set of brain regions as a graph, with constant structure and time-varying signals. GFTs were computed for BOLD signals and their coefficients were merged into a connectivity matrix. This method makes the determination of discriminative graph frequency patterns feasible. Such features were used by a decision tree to separate ASD from neurotypical subjects. The resulting framework was more accurate than other models based on SC or FC.

B. Filtering

Another study [2] presented an anisotropic spatial filtering scheme for fMRI data in white matter (WM). Using diffusion orientation distribution functions (ODF), the authors built subject-specific voxel-wise WM graphs, on whose spectrums heat kernel filters were defined. The resulting filters adapt to the local axonal orientation when applied at any position within the WM, smoothing fMRI data. This scheme exhibited greater sensitivity and specificity for detection of slender anisotropic activations, when compared to isotropic Gaussian filters.

C. Situation Awareness

A spectral-spatial model was proposed [3] for fatigue-related Situation Awareness (SA) recognition. The topological structure of Electroencephalogram (EEG) channels was described by an adjacency matrix based on the Euclidian distance between channels. Graph convolution was used to perform neighborhood aggregation, resulting in extraction of spatial features. This model achieved greater accuracy on SA recognition from EEG signals than traditional machine learning and deep learning methods.

D. Computer Vision

In computer vision, GSP may be used to subtract backgrounds [4]. The GraphBGS (Graph BackGround Subtraction) algorithm uses Cascade Mask Region Convolutional Neural

Network (R-CNN) for instance segmentation and a temporal median filter to initialize backgrounds. Each instance is modelled as a node of a graph and is represented by optical flow, intensity and texture features. After the graph is built, some nodes are sampled and a semi-supervised reconstruction algorithm classifies an instance as a moving or static object. This algorithm is adaptive to complex scenarios and outperforms state-of-the-art methods.

E. Image Processing

In image processing, GSP may be used to classify image textures in a method based on graph wavelet transforms [5]. A texture is treated as a graph signal, which is decomposed into multiscale components by graph wavelet filter banks. Singular value decomposition (SVD) is applied to each resulting subband. To increase tolerance to noise, the maximum, mean and median values of each subband are extracted and modelled with Weibull distributions. The output parameters are used for texture classification.

F. Swarm Intelligence

GSP is an important resource in the detection of anomalous swarming agents [6]. Each agent is defined as a vertex in a graph and an adjacency matrix is built for each instance in time. Depending on the scenario, different graph signals are adopted, such as normalized velocity, adjusted position and exponential signals. GFT is used to create spectral concentration in a small set of low frequency harmonics. A group of several snapshots of swarming data may be decisive for an effective detection.

G. Electrical Engineering

In electrical engineering, a parametrized and scalable graph neural network (GNN) may be trained to find an approximation to the optimal power flow (OPF) in a grid [7]. The problem of OPF is to determine the power each generator has to provide to the grid in order to meet demands, with minimal production costs. The main approach is to use a nonlinear map consisting of a cascade of L layers, each of them applying a graph convolution and a pointwise nonlinearity, to imitate the optimal solution. Global and local GNNs had greater performances when compared to global and local multilayer perceptrons (MLP).

VI. CONCLUSIONS

In this paper, we reviewed theoretical concepts of graph signal processing. Mathematical properties were adapted from traditional signal processing, allowing manipulation and filtering of signals in flexible graph structures. Many scientific publications are based on recently proposed methods, making GSP an emerging field of research with a wide range of applications, which may require different tools and specific approaches.

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